Software Requirements Specification (SRS)

Project: Insurance Workflow Automation System
Version: 1.0
Authors: Sanjith Suresh Kumar (Lead), Sakaray Vishal, Samarth A, Sampreeth Jaladi
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Revision History

Version	Date	Author	Change Summary	Approval
1.0	18-08-2025	Sanjith Suresh Kumar	SRS with UML diagrams and RTM embedded	Approved
0.9	10-08-2025	Sakaray Vishal	Initial draft — functional requirements	Reviewed

Approvals

Role	Name	Signature / Email	Date
Project Lead	Sanjith Suresh Kumar	sanjith.sk@pes.edu	18-08-2025
Course Coordinator	—	—	—

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the **Insurance Workflow Automation System**. It defines functional and non-functional requirements, interfaces, and verification criteria for automating the end-to-end lifecycle of insurance policy processing — from quote generation and underwriting to claim settlement.

1.2 Scope

The system covers:

- Customer onboarding and policy quote generation
- Automated underwriting and risk assessment
- Policy issuance, renewal, and endorsement
- Claim registration, adjudication, and settlement
- Agent / broker portal and admin dashboard
- Workflow orchestration engine (BPMN-based)

Excludes: Core reinsurance internals, third-party actuarial model implementation, and external banking settlement networks (integrated via APIs only).

1.3 Audience

Developers, QA Engineers, System Integrators, Insurance Operations Staff, Compliance Auditors, and Assessment Evaluators.

1.4 Definitions

Acronyms: IWAS (Insurance Workflow Automation System), KYC (Know Your Customer), BPMN (Business Process Model and Notation), RTM (Requirements Traceability Matrix), TLS, API, UI, CI/CD, SLA.

2. Overall Description

2.1 Product Perspective

The IWAS is a cloud-native, microservices-based platform that orchestrates insurance business workflows. It integrates with KYC providers, payment gateways, actuarial engines, and existing policy administration systems via REST APIs. It exposes a web portal for customers, agents, and internal staff.

2.2 Major Product Functions

- Generate insurance quotes based on user profile and risk factors
- Automated underwriting with rule engine and risk scoring
- Issue, renew, and endorse policies
- Register and track claims with document upload
- Route claims through multi-level approval workflows
- Trigger notifications (email, SMS, in-app)
- Generate reports and dashboards for operations
- Audit logging and compliance tracking

2.3 User Roles and Characteristics

- **Customer:** Buys policies, files claims, tracks status. Expects simple UI and quick turnaround.
- **Agent / Broker:** Onboards customers, manages portfolios, needs bulk operations.
- **Underwriter:** Reviews risk, approves/rejects policies. Needs detailed customer data and rule-based flags.
- **Claims Adjuster:** Reviews claims, approves settlements. Needs fraud indicators and document verification.
- **Admin / Ops:** Configures workflows, manages users, monitors SLAs.

2.4 Operating Environment

Cloud-hosted (AWS/Azure), containerized (Docker + Kubernetes), web-based UI (Chrome, Firefox, Edge), REST APIs, PostgreSQL database, Redis cache, Kafka message broker.

2.5 Constraints

IRDAI compliance, GDPR / data privacy, PCI-DSS for premium payments, TLS 1.2+ mandatory, BPMN 2.0 workflow standards.

3. External Interface Requirements

3.1 User Interfaces

Responsive web UI (React). Role-based dashboards. WCAG 2.1 AA accessibility. Multi-language support (English + regional).

3.2 Hardware Interfaces

- Server infrastructure (cloud VMs / Kubernetes nodes)
- HSM for cryptographic key storage
- Standard client devices (desktop, tablet, mobile browser)

3.3 Software Interfaces

- **KYC Provider API** — Aadhaar / PAN verification
- **Payment Gateway** — Premium collection (Razorpay/Stripe)
- **Actuarial Engine** — Premium calculation
- **Notification Service** — Email (SendGrid), SMS (Twilio)
- **Document Storage** — S3-compatible object store
- **Legacy Policy Admin System** — SOAP/REST bridge

3.4 Communications

TLS 1.2+ enforced, mutual TLS for internal services, JWT-based auth, OAuth 2.0 for third-party integrations, retry with exponential backoff.

4. System Features (Detailed)

4.1 Quote Generation & Underwriting

Req ID	Requirement	Type	Priority	Source	Acceptance Criteria / Test	Comments
IWA-F-001	The system shall generate a personalized insurance quote based on age, income, and coverage type within 10 seconds.	Functional	High	Business	AC-IWA-F-001: Quote returned with premium breakdown. Test: TC-Quote-01	Requires actuarial API
IWA-F-002	The system shall perform automated risk scoring and flag high-risk applications for manual review.	Functional	High	Underwriting	AC-IWA-F-002: Risk score 0–100 with threshold config. Test: TC-UW-01	Rule engine
IWA-F-003	The system shall verify customer KYC via integrated KYC API before policy issuance.	Functional	High	Compliance	AC-IWA-F-003: KYC pass/fail returned. Test: TC-KYC-01	Third-party API

4.2 Policy Issuance & Management

Req ID	Requirement	Type	Priority	Source	Acceptance Criteria / Test	Comments
IWA-F-010	The system shall issue a digital policy document (PDF) upon successful premium payment.	Functional	High	Business	AC-IWA-F-010: Signed PDF with policy number. Test: TC-Policy-01	Digital signature
IWA-F-011	The system shall support policy renewal with automated reminders 30, 15, and 7 days before expiry.	Functional	High	Operations	AC-IWA-F-011: Notifications sent. Test: TC-Renew-01	Notification service
IWA-F-012	The system shall allow endorsement (modification) of active policies with audit trail.	Functional	Medium	Operations	AC-IWA-F-012: Endorsement recorded. Test: TC-Endorse-01	—
IWA-F-013	The system shall generate and store invoices for premium payments.	Functional	Medium	Finance	AC-IWA-F-013: Invoice PDF with tax breakup. Test: TC-Invoice-01	—

4.3 Claims Processing

Req ID	Requirement	Type	Priority	Source	Acceptance Criteria / Test	Comments
IWA-F-020	The system shall allow customers to register claims with supporting document upload.	Functional	High	Business	AC-IWA-F-020: Claim ID generated. Test: TC-Claim-01	File size limit 10MB
IWA-F-021	The system shall route claims through configurable multi-level approval workflows (BPMN).	Functional	High	Operations	AC-IWA-F-021: Workflow executed per config. Test: TC-WF-01	BPMN engine
IWA-F-022	The system shall run automated fraud detection checks (duplicate claims, unusual patterns).	Functional	High	Risk	AC-IWA-F-022: Fraud flag raised. Test: TC-Fraud-01	Rule-based
IWA-F-023	The system shall calculate claim settlement amount based on policy terms.	Functional	High	Finance	AC-IWA-F-023: Settlement matches formula. Test: TC-Settle-01	—
IWA-F-024	The system shall notify the claimant at every status change (registered, under review, approved, settled).	Functional	Medium	Customer	AC-IWA-F-024: Notifications received. Test: TC-Notify-01	—

4.4 Workflow Orchestration & Admin

Req ID	Requirement	Type	Priority	Source	Acceptance Criteria / Test	Comments
IWA-F-030	The system shall provide an admin UI to design and deploy BPMN workflows.	Functional	High	Operations	AC-IWA-F-030: Workflow deployed. Test: TC-Admin-01	—
IWA-F-031	The system shall support role-based access control (RBAC) for all modules.	Functional	High	Security	AC-IWA-F-031: Unauthorized access denied. Test: TC-RBAC-01	—
IWA-F-032	The system shall provide dashboards with SLA metrics, claim backlog, and policy counts.	Functional	Medium	Management	AC-IWA-F-032: Dashboard loads ≤3s. Test: TC-Dash-01	—
IWA-F-033	The system shall export reports in CSV and PDF formats.	Functional	Low	Operations	AC-IWA-F-033: Export valid. Test: TC-Report-01	—
IWA-F-034	The system shall maintain an immutable audit log of all workflow actions.	Functional	High	Compliance	AC-IWA-F-034: Audit log tamper-proof. Test: TC-Audit-01	Append-only

5. Non-Functional Requirements (Detailed)

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
IWA-NF-001	Quote generation and claim status APIs shall respond within 2 seconds for 95% of requests under normal load.	Performance	High	95th percentile ≤ 2s. Test: TC-Perf-01
IWA-NF-002	System shall provide 99.9% availability monthly (excluding scheduled maintenance).	Reliability	High	Uptime ≥99.9%. Test: Ops reports
IWA-NF-003	All sensitive customer data (PII, policy data) must be encrypted at rest (AES-256) and in transit (TLS 1.2+).	Security	High	Encryption verified. Test: TC-Sec-01
IWA-NF-004	System shall retain audit logs for 7 years with immutable storage.	Audit/ Compliance	High	Retention policy verified. Test: TC-OPS-01
IWA-NF-005	Web UI shall conform to WCAG 2.1 AA accessibility standards.	Usability	Medium	Accessibility audit pass. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

1. Protect customer PII and policy data from unauthorized access and disclosure.
2. Ensure integrity and non-repudiation of all financial and claim transactions.
3. Achieve and maintain compliance with IRDAI, GDPR, and PCI-DSS standards.

5.1.2 Security Requirements

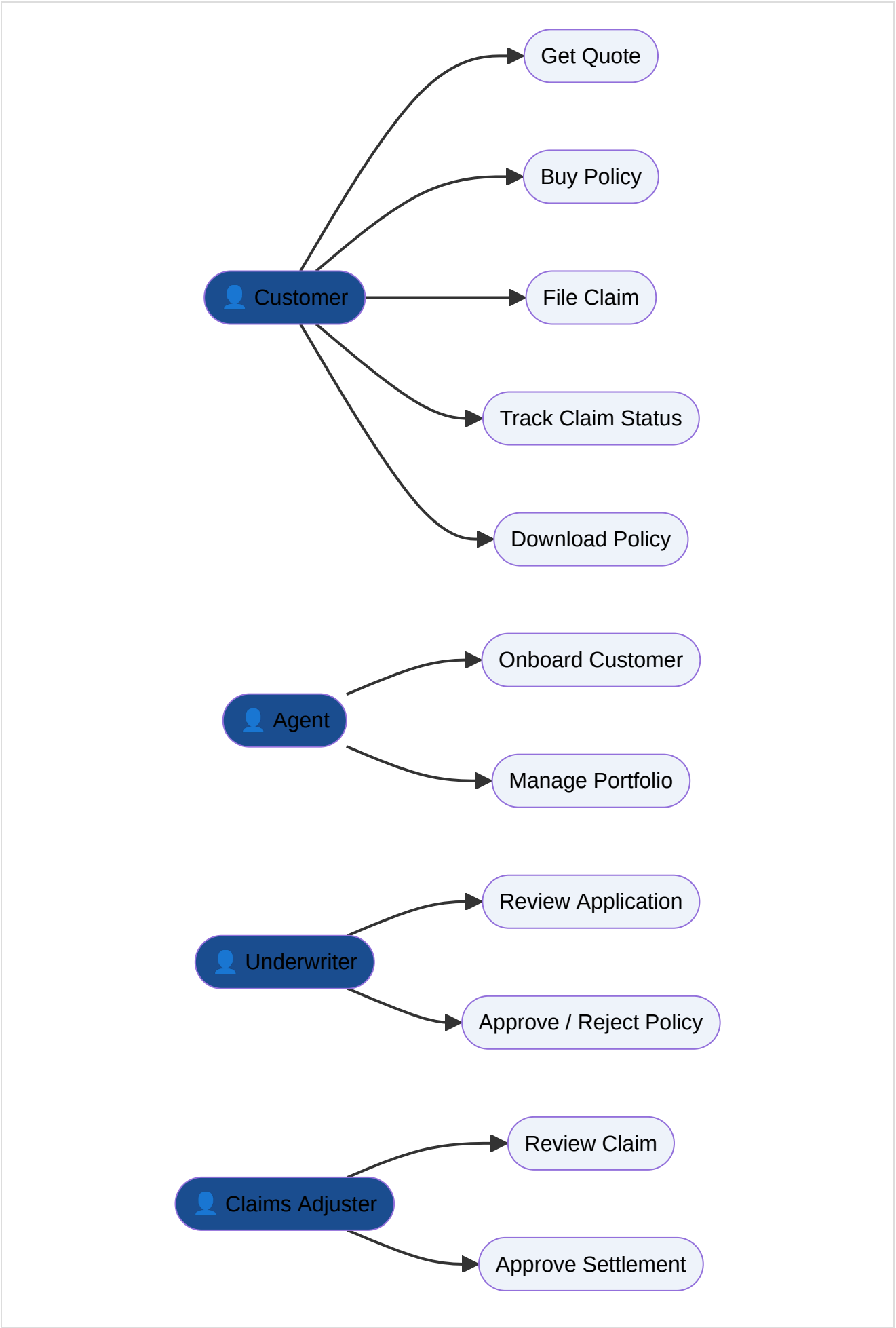
Req ID	Requirement	Type	Priority	Acceptance Criteria / Test
IWA-SR-001	TLS 1.2+ mandatory for all network connections.	Security	High	SSL scan pass. Test: TC-Sec-TLS-01
IWA-SR-002	The system shall enforce multi-factor authentication (MFA) for admin and underwriter roles.	Security	High	MFA enforced. Test: TC-Sec-MFA-01
IWA-SR-003	The system shall mask PII in logs and UI except for authorized roles.	Security	High	PII masked. Test: TC-Sec-PII-01
IWA-SR-004	The system shall implement rate limiting and IP throttling on public APIs.	Security	High	Rate limit triggers. Test: TC-Sec-RL-01
IWA-SR-005	The system shall digitally sign all issued policy and claim settlement documents.	Security	High	Signature valid. Test: TC-Sec-Sign-01

6. Quality Attributes & Acceptance Tests

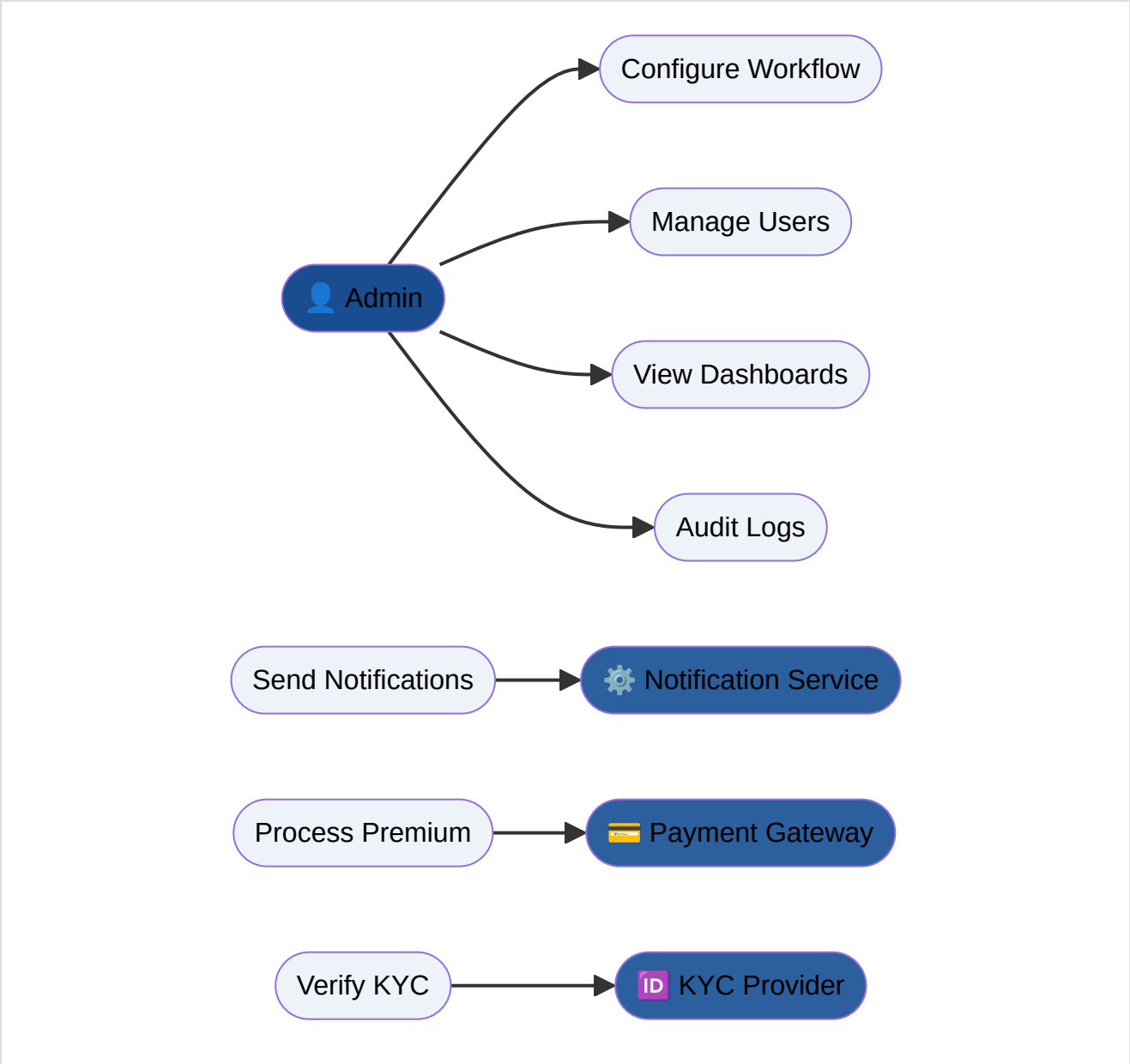
- **Exit criteria for acceptance:** All high-priority functional requirements implemented and verified; no critical NFR failures; RTM shows 100% of high-priority test cases passed.
- **Acceptance test suites:** Quote Generation, Underwriting, Policy Issuance, Claims, Workflow, Performance, Security, Accessibility.

7. System Models and Diagrams

7.1 UML Use-Case Diagram — Customer & Agent



7.2 UML Use-Case Diagram — Admin & External Systems



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
IWA-F-001	Generate Quote	4.1 / DS-Quote-01	QuoteModule	TC-Quote-01	N	—
IWA-F-002	Risk Scoring	4.1 / DS-UW-01	UnderwritingModule	TC-UW-01	N	—
IWA-F-010	Issue Policy	4.2 / DS-Policy-01	PolicyModule	TC-Policy-01	N	—
IWA-F-020	Register Claim	4.3 / DS-Claim-01	ClaimsModule	TC-Claim-01	N	—
IWA-F-021	BPMN Workflow Routing	4.3 / DS-WF-01	WorkflowEngine	TC-WF-01	N	—
IWA-NF-001	Response Time ≤ 2s	5 / DS-Perf-01	WebUI / CoreAPI	TC-Perf-01	N	—
IWA-SR-001	TLS 1.2+	5.1.2 / DS-Sec-01	Gateway	TC-Sec-TLS-01	N	—